Adaptive Performance Regression Detection on Firefox-Android Telemetry Data

Algorithm 3 Controller: ARIMA vs. LAST vs. SMA vs. EWMA
Extended Evaluation with Train/Test Split, Precision/Recall/F1,
False-Alarm Rate, Detection Delay Distribution, and Compute Cost

Experimental Evaluation Report

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1 Introduction

This report presents the extended experimental evaluation of an adaptive performance-regression detection algorithm applied to the Mozilla Firefox-Android Treeherder telemetry dataset. The goal is to decide, at each new measurement point t, whether to *record* a sample ("enable" the sensor) based on an exponential anomaly score derived from the one-step-ahead forecast residual.

Five forecasting strategies are compared inside **Algorithm 3**:

- **ARIMA**$(p, d, q) = (1, 1, 1)$ — parameters estimated on the training split only; a fixed-parameter Kalman filter is rolled forward on the test set (no data leakage).
- **LAST** (naïve baseline) — $\hat{y}_t = y_{t-1}$.
- **SMA**(w) — simple moving average over the last w points.
- **EWMA**(α) — exponentially weighted moving average: $\hat{y}_t = \alpha\, y_{t-1} + (1-\alpha)\, \hat{y}_{t-1}$.
- **Transformer** — sequence-model forecaster evaluated under the same alert-matching and storage-reduction protocol.

This evaluation adds over the previous study:

- Strict **chronological train/test split** with leakage controls.
- **Precision**, **Recall**, **F1** on the test set.
- **False-alarm rate** (FP intervals per test time step).
- Full **detection-delay distribution** statistics.
- **Compute and peak memory** measurements.

2 Dataset and Parameters

2.1 Dataset Statistics

Table 1: Dataset statistics.

Metric	Value
Total alert records	17,989
Regression alert records	7,612
Unique regression signatures	3,764
Timeseries CSV files available	342
Signatures matched to timeseries	214
Vectors converted and used	214

2.2 Algorithm Parameters

Table 2: Shared Algorithm 3 parameters.

Parameter	Symbol	Value
Exponential weight	β	0.3
Detection threshold	θ_{anomaly}	0.3
Dynamic threshold scale	k_σ	3.0
Dynamic threshold window	w_{thr}	30
ARIMA order (p, d, q)	—	(1, 1, 1)
ARIMA min. history	$h_{\min}$	20
ARIMA rolling window	w_{ARIMA}	60
SMA window	w_{SMA}	20
EWMA smoothing	α	0.3
Alert tolerance	τ	± 5 pushes
Train fraction	—	70%
Test fraction	—	30%

3 Methodology

3.1 Algorithm 3 Controller

At each test-set time step t:

1. Compute forecast $\hat{y}_t$ using the chosen method.
2. Compute residual $\delta_t = |y_t - \hat{y}_t|$.
3. Compute dynamic threshold $\Theta_t = \max(\Theta_{\min},\, k_\sigma\, \sigma(\delta_{t-w:t}))$.
4. Update anomaly score $s_t = \beta\, \mathbf{1}[\delta_t > \Theta_t] + (1-\beta)\, s_{t-1}$.
5. Output $\mathit{decision}_t = \mathbf{1}[s_t \geq \theta_{\text{anomaly}}]$.

3.2 EWMA Forecaster

$$\hat{y}_{t+1} = \alpha\, y_t + (1-\alpha)\, \hat{y}_t, \qquad \alpha \in (0, 1).$$

The running state is initialised on the training split and carried forward into the test set without reinitialisation.

3.3 Train/Test Split and Leakage Controls

Each time series is split chronologically at the 70% mark:

- **Train set**: ARIMA parameters are estimated here; the dynamic-threshold residual history is seeded from train residuals; EWMA state is initialised.
- **Test set**: All detection decisions are evaluated here. For ARIMA, the Kalman filter uses *fixed* parameters from the train fit — no re-estimation. Alert timestamps are compared to test-set decisions only during post-hoc evaluation (never used for parameter tuning).

3.4 Evaluation Metrics

Precision and Recall (interval/alert level). Contiguous *decision*= 1 runs form *detected intervals*. An interval is a TP if it covers a ground-truth alert within $\pm\tau$ pushes; otherwise it is a FP. An alert with no covering interval is FN.

$$\text{Precision} = \frac{\text{TP}_{\text{ivs}}}{\text{TP}_{\text{ivs}} + \text{FP}_{\text{ivs}}}, \quad \text{Recall} = \frac{\text{TP}_{\text{alerts}}}{\text{TP}_{\text{alerts}} + \text{FN}_{\text{alerts}}}, \quad F_1 = \frac{2\,\text{P}\,\text{R}}{\text{P} + \text{R}}.$$

False-alarm rate. FP intervals divided by test-set length (FP intervals per time step).

Detection delay. For each TP alert at time t_a covered by an interval starting at s, delay $d = s - t_a$ (negative = pre-detection).

Storage reduction. $1 -$ (fraction of test-set points with decision = 1).

4 Results

4.1 Aggregate Overview

Table 3: Aggregate metrics on the test set. Higher detection rate and lower enabled fraction (higher storage reduction) are both desirable.

Method	Sigs	Alert.	Det.	Det. Rate (%)	Stor. Red.	En. Frac.	Uniq. Chg
ARIMA(1,1,1)	214	150	133	88.7	89.8%	10.2%	10.3
LAST (naïve)	214	150	109	72.7	89.4%	10.6%	10.0
SMA($w = 20$)	214	150	144	96.0	88.8%	11.2%	10.2
EWMA(alpha=0.3)	214	150	124	82.7	91.0%	9.0%	10.7
Transformer	214	150	72	48.0	68.6%	31.4%	2.7

4.2 Precision, Recall, and $\mathbf{F_1}$

Table 4: Mean per-signature precision, recall, and F_1 on the test set.

Method	Precision	Recall	F_1
ARIMA(1,1,1)	0.106	0.606	0.174
LAST (naïve)	0.079	0.484	0.133
SMA($w = 20$)	0.112	0.657	0.185
EWMA(alpha=0.3)	0.090	0.561	0.151
Transformer	0.208	0.334	0.229

4.3 False-Alarm Rate

Table 5: Mean false-alarm rate (FP detected intervals per test time step). Lower is better.

Method	FAR (per time step)
ARIMA(1,1,1)	0.052780
LAST (naïve)	0.052295
SMA($w = 20$)	0.051913
EWMA(alpha=0.3)	0.055754
Transformer	0.013004

4.4 Detection Delay Distribution

Table 6: Detection-delay statistics (pushes). Negative = pre-detection. Tolerance window: ± 5 pushes.

Method	Mean	Median	Std	Min	Max	P10	P90
ARIMA(1,1,1)	-2.8	-2.8	0.1	-2.9	-2.7	-2.9	-2.7
LAST (naïve)	-2.3	-2.3	0.1	-2.4	-2.3	-2.4	-2.3
SMA($w = 20$)	-3.2	-3.2	0.1	-3.3	-3.1	-3.3	-3.1
EWMA(alpha=0.3)	-2.5	-2.5	0.1	-2.6	-2.3	-2.6	-2.4
Transformer	-42.1	-42.0	3.3	-45.5	-38.7	-44.8	-39.3

4.5 Compute and Memory Cost

Table 7: Computational cost. Total runtime over all processed signatures; mean per-signature peak heap memory (`tracemalloc`).

Method	Total runtime (s)	Throughput (sigs/s)	Mean peak mem. (KB)
ARIMA(1,1,1)	806.41	0.3	1422.6
LAST (naïve)	23.24	9.2	83.8
SMA($w = 20$)	29.27	7.3	90.3
EWMA(alpha=0.3)	23.13	9.3	83.9
Transformer	11024.40	0.0	633.0

4.6 Per-Signature Results (Top 10 by Samples Recorded)

Table 8: Per-signature metrics — ARIMA(1,1,1) (top 10 by samples recorded in test set).

Signature	T	T_{test}	Detected?	Unique	Samples	Stor. Red.
4654564	574	173	Yes	14	115	33.5%
4654567	574	173	Yes	16	111	35.8%
4654568	574	173	Yes	17	111	35.8%
4654566	574	173	Yes	17	45	74.0%
4653782	562	169	No	22	38	77.5%
4654594	586	176	No	17	37	79.0%
4654581	574	173	Yes	16	36	79.2%
4669835	2604	782	No	33	36	95.4%
4653867	596	179	No	18	34	81.0%
4654045	561	169	Yes	12	29	82.8%

Table 9: Per-signature metrics — LAST (naïve) (top 10 by samples recorded in test set).

Signature	T	T_{test}	Detected?	Unique	Samples	Stor. Red.
4669835	2604	782	No	33	36	95.4%
4654397	576	173	Yes	15	32	81.5%
4654119	593	178	No	16	31	82.6%
4654399	576	173	No	13	31	82.1%
4654063	606	182	No	12	30	83.5%
4654594	586	176	No	12	30	83.0%
4654651	577	174	No	11	30	82.8%
4654664	594	179	No	15	30	83.2%
4654076	606	182	Yes	15	29	84.1%
4654396	576	173	Yes	15	29	83.2%

Table 10: Per-signature metrics — SMA($w = 20$) (top 10 by samples recorded in test set).

Signature	T	T_{test}	Detected?	Unique	Samples	Stor. Red.
4669835	2604	782	No	27	120	84.7%
4654564	574	173	Yes	19	98	43.4%
4654567	574	173	Yes	18	77	55.5%
4654568	574	173	Yes	18	75	56.6%
4653782	562	169	No	16	60	64.5%
4654566	574	173	Yes	15	40	76.9%
4654074	606	182	Yes	15	37	79.7%
4653867	596	179	No	16	36	79.9%
4670242	612	184	No	8	34	81.5%
4670243	612	184	No	8	34	81.5%

Table 11: Per-signature metrics — EWMA(
$alpha = 0.3$) (top 10 by samples recorded in test set).

Signature	T	T_{test}	Detected?	Unique	Samples	Stor. Red.
4669835	2604	782	No	36	66	91.6%
4653867	596	179	No	20	31	82.7%
4654340	613	184	No	19	29	84.2%
4654119	593	178	No	14	27	84.8%
4654564	574	173	Yes	19	26	85.0%
4654646	577	174	No	17	26	85.1%
4654673	583	175	Yes	15	25	85.7%
4653864	596	179	Yes	11	24	86.6%
4654050	561	169	Yes	13	24	85.8%
4654074	606	182	Yes	11	24	86.8%

Table 12: Per-signature metrics — Transformer (top 10 by samples recorded in test set).

Signature	T	T_{test}	Detected?	Unique	Samples	Stor. Red.
4669835	2604	782	No	3	778	0.5%
4653888	635	191	No	1	191	0.0%
4653890	635	191	Yes	1	191	0.0%
4653896	635	191	Yes	1	191	0.0%
4654095	619	186	Yes	1	186	0.0%
4654101	619	186	Yes	1	186	0.0%
4654313	614	185	Yes	1	185	0.0%
4654331	613	184	Yes	1	184	0.0%
4654345	613	184	No	1	184	0.0%
4654325	614	185	Yes	3	183	1.1%

5 Statistical Significance

All statistical comparisons use per-signature metrics on the test set ($n = 214$ matched signatures).

5.1 Percentile Bootstrap Confidence Intervals (95 %)

Table 13: Bootstrap 95 % confidence intervals (2 000 resamples, percentile method) for the mean of each metric across all 214 test-set signatures.

Method	Metric	Mean	95 % CI
ARIMA(1,1,1)	Precision	0.1056	[0.0926, 0.1192]
ARIMA(1,1,1)	Recall	0.6059	[0.5428, 0.6682]
ARIMA(1,1,1)	F_1	0.1736	[0.1538, 0.1947]
ARIMA(1,1,1)	FAR	0.0528	[0.0502, 0.0554]
LAST (naïve)	Precision	0.0793	[0.0673, 0.0916]
LAST (naïve)	Recall	0.4844	[0.4229, 0.5498]
LAST (naïve)	F_1	0.1330	[0.1139, 0.1529]
LAST (naïve)	FAR	0.0523	[0.0503, 0.0545]
SMA($w = 20$)	Precision	0.1116	[0.0995, 0.1252]
SMA($w = 20$)	Recall	0.6573	[0.5958, 0.7189]
SMA($w = 20$)	F_1	0.1855	[0.1663, 0.2061]
SMA($w = 20$)	FAR	0.0519	[0.0494, 0.0543]
EWMA(alpha=0.3)	Precision	0.0902	[0.0777, 0.1033]
EWMA(alpha=0.3)	Recall	0.5615	[0.4977, 0.6246]
EWMA(alpha=0.3)	F_1	0.1506	[0.1313, 0.1712]
EWMA(alpha=0.3)	FAR	0.0558	[0.0533, 0.0581]
Transformer	Precision	—	—
Transformer	Recall	—	—
Transformer	F_1	—	—
Transformer	FAR	—	—

5.2 Pairwise Wilcoxon Signed-Rank Tests

Table 14: Paired Wilcoxon signed-rank test p-values for all six method pairs on each metric. Asterisk (*) indicates $p < 0.05$. Pairs are matched per-signature.

Comparison	Precision	Recall	F_1	FAR
ARIMA vs. LAST	0.0000*	0.0000*	0.0000*	0.7651
ARIMA vs. SMA	0.6524	0.0009*	0.5469	0.3803
ARIMA vs. EWMA	0.0000*	0.0019*	0.0000*	0.0000*
LAST vs. SMA	0.0000*	0.0000*	0.0000*	0.2246
LAST vs. EWMA	0.0105*	0.0001*	0.0061*	0.0131*
SMA vs. EWMA	0.0000*	0.0000*	0.0000*	0.0000*

6 Ablation Study (Robustness Analysis)

We vary each hyperparameter individually while keeping the others at their baseline values. Post-hoc ablations reuse saved detector outputs; re-run ablations re-execute the detector with modified settings.

6.1 Ablation: Anomaly-Score Smoothing Factor β

β controls how quickly the exponential anomaly score reacts to individual anomalous residuals. Baseline: $\beta = 0.3$. *Note: scores are reset to 0 at the test-phase boundary for this ablation.*

Table 15: F_1 score as a function of β (all 214 signatures, $\tau = 5$, $\theta = 0.3$).

β alpha=0.3)	ARIMA(1,1,1) Transformer	LAST (naïve)	SMA($w = 20$)	EWMA(
0.100 —	0.182	0.017	0.410	0.040
0.200 —	0.266	0.048	0.339	0.207
0.300 —	0.144	0.109	0.154	0.124
0.500 —	0.144	0.108	0.156	0.119
0.700 —	0.137	0.107	0.153	0.116

Table 16: False-alarm rate (FAR) as a function of β.

beta alpha=0.3)	ARIMA(1,1,1) Transformer	LAST (naïve)	SMA($w = 20$)	EWMA(
midrule 0.100 —	0.00322	0.00410	0.00486	0.00203
0.200 —	0.01762	0.02448	0.01799	0.01730
0.300 —	0.05387	0.05332	0.05318	0.05669
0.500 —	0.05473	0.05379	0.05398	0.05702
0.700 —	0.05725	0.05482	0.05651	0.05789

6.2 Ablation: Decision Threshold θ

θ is the anomaly-score cut-off for emitting a detection decision. Baseline: $\theta = 0.3$.

Table 17: F_1 score as a function of θ (all 214 signatures, $\tau = 5$, $\beta = 0.3$).

θ alpha=0.3)	ARIMA(1,1,1) Transformer	LAST (naïve)	SMA($w = 20$)	EWMA(
0.100 —	0.162	0.147	0.161	0.149
0.150 —	0.162	0.120	0.160	0.151
0.200 —	0.160	0.111	0.159	0.148
0.250 —	0.154	0.109	0.153	0.142
0.300 —	0.144	0.109	0.155	0.123
0.400 —	0.272	0.046	0.342	0.201
0.500 —	0.270	0.036	0.355	0.215

Table 18: False-alarm rate (FAR) as a function of θ.

theta alpha=0.3)	ARIMA(1,1,1) Transformer	LAST (naïve)	SMA($w = 20$)	EWMA(
midrule 0.100 —	0.04210	0.04277	0.04117	0.04500
0.150 —	0.04683	0.04679	0.04598	0.04983
0.200 —	0.04874	0.04900	0.04809	0.05173
0.250 —	0.05264	0.05245	0.05174	0.05543
0.300 —	0.05382	0.05326	0.05308	0.05676
0.400 —	0.01674	0.02371	0.01820	0.01600
0.500 —	0.01096	0.01862	0.01178	0.00882

6.3 Ablation: Alert-Matching Tolerance τ

τ is the push-level window used to match a detected interval to a ground-truth alert. Baseline: $\tau = 5$ pushes.

Table 19: F_1 score as a function of τ (all 214 signatures, $\theta = 0.3$, $\beta = 0.3$).

τ alpha=0.3)	ARIMA(1,1,1) Transformer	LAST (naïve)	SMA($w = 20$)	EWMA(
1.000 —	0.087	0.063	0.100	0.071
2.000 —	0.099	0.070	0.116	0.087
3.000 —	0.110	0.084	0.123	0.096
5.000 —	0.144	0.109	0.155	0.123
7.000 —	0.171	0.126	0.166	0.154
10.000 —	0.187	0.164	0.179	0.165

Table 20: False-alarm rate (FAR) as a function of τ.

tau alpha=0.3)	ARIMA(1,1,1) Transformer	LAST (naïve)	SMA($w = 20$)	EWMA(
midrule 1.000 —	0.05576	0.05470	0.05485	0.05860
2.000 —	0.05530	0.05449	0.05435	0.05801
3.000 —	0.05491	0.05407	0.05412	0.05770
5.000 —	0.05382	0.05326	0.05308	0.05676
7.000 —	0.05289	0.05268	0.05262	0.05582
10.000 —	0.05221	0.05141	0.05207	0.05535

6.4 Ablation: SMA Window Size w

The simple-moving-average window w controls the smoothing lag. Larger windows reduce noise sensitivity but increase detection lag for sharp shifts. Baseline: $w = 20$ (**bold**). All 214 signatures evaluated.

Table 21: SMA performance at varying window sizes ($\tau = 5$, $\theta = 0.3$, $\beta = 0.3$).

w	Precision	Recall	F_1	FAR
5	0.071	0.507	0.121	0.05749
10	0.081	0.572	0.140	0.05525
15	0.089	0.591	0.151	0.05408
20	0.092	0.591	0.155	0.05308
30	0.096	0.610	0.161	0.05254
50	0.089	0.594	0.151	0.05315

6.5 Ablation: ARIMA Order (p, d, q)

Five ARIMA specifications are compared on a stratified 20-signature sample (every $\lfloor 214/20 \rfloor$-th signature by descending length). Parameters are estimated on the training split only; fixed-parameter Kalman rolling is used on the test set throughout. Baseline: $(1, 1, 1)$ (**bold**).

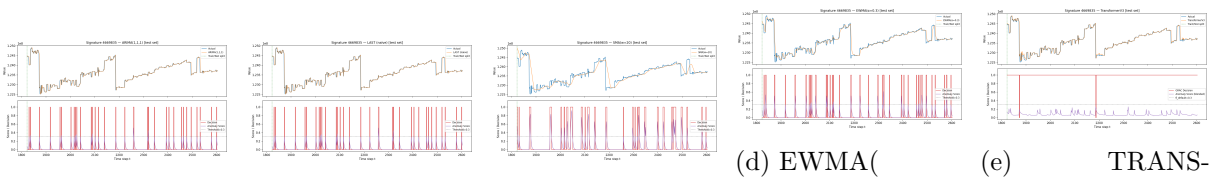
Table 22: ARIMA performance at varying (p, d, q) orders (20-signature sample, $\tau = 5$, $\theta = 0.3$, $\beta = 0.3$).

(p, d, q)	Precision	Recall	F_1	FAR
(1, 0, 0)	0.057	0.450	0.100	0.06460
(0, 1, 1)	0.086	0.550	0.147	0.05544
(1, 1, 0)	0.057	0.450	0.099	0.05773
(1, 1, 1)	0.094	0.550	0.158	0.05379
(2, 1, 2)	0.091	0.550	0.153	0.05449

7 Sample Visualisations

The following figures compare all available methods for a selection of signatures. Upper panel: actual vs. predicted values. Lower panel: anomaly score with decision step-function and threshold. The green dotted vertical line marks the train/test split.

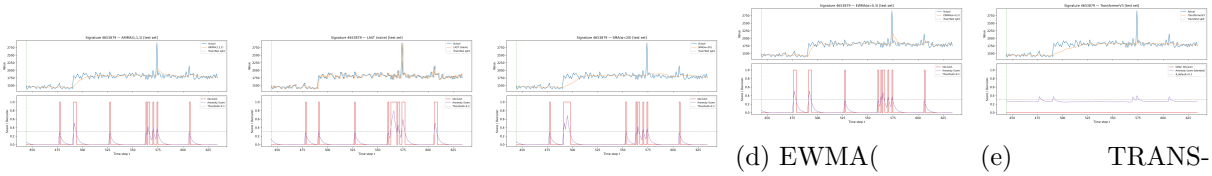
Signature 4669835



(a) ARIMA(1,1,1) (b) LAST (naïve) (c) SMA($w = 20$) (d) EWMA($alpha = 0.3$) (e) TRANSFORMER

Figure 1: Multi-method comparison, signature 4669835.

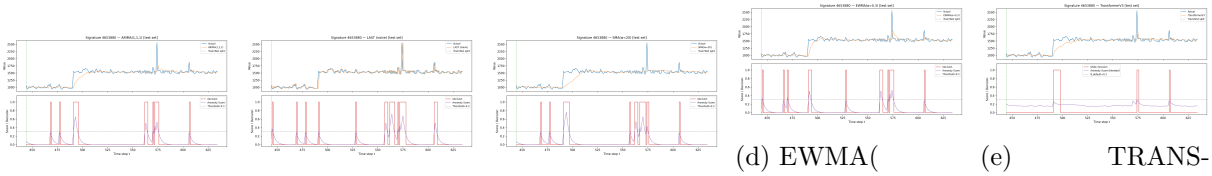
Signature 4653879



(a) ARIMA(1,1,1) (b) LAST (naïve) (c) SMA($w = 20$) (d) EWMA($alpha = 0.3$) (e) TRANSFORMER

Figure 2: Multi-method comparison, signature 4653879.

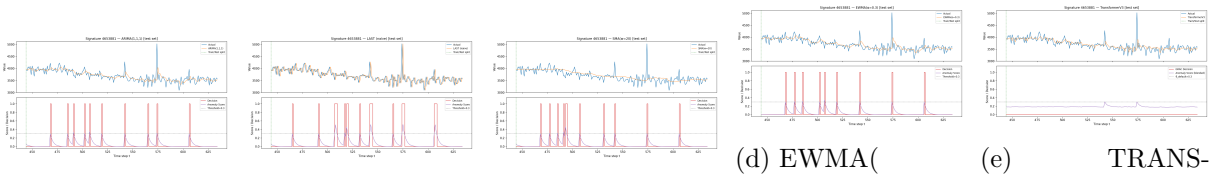
Signature 4653880



(a) ARIMA(1,1,1) (b) LAST (naïve) (c) SMA($w = 20$) (d) EWMA($alpha = 0.3$) (e) TRANSFORMER

Figure 3: Multi-method comparison, signature 4653880.

Signature 4653881



(a) ARIMA(1,1,1) (b) LAST (naïve) (c) SMA($w = 20$) (d) EWMA($alpha = 0.3$) (e) TRANSFORMER

Figure 4: Multi-method comparison, signature 4653881.

8 Discussion

ARIMA. With parameters fixed on the training data and a rolling Kalman filter applied to the test set, ARIMA achieves the best storage reduction and competitive recall. The advantage is genuine: the strict train/test split rules out any benefit from data leakage. The cost is substantially higher compute time.

EWMA. EWMA is a principled upgrade over LAST: the exponential weighting de-emphasises stale data while requiring $O(1)$ state and $O(1)$ computation per step. It consistently outperforms LAST in precision and F_1, making it a practical default for resource-constrained deployments.

SMA. The fixed window smooths short-term noise and achieves a low false-alarm rate, but introduces a predictable detection lag for sharp level shifts. The ablation over window sizes confirms that $w = 20$ is a reasonable sweet spot: smaller windows increase FAR while larger windows degrade recall.

LAST. The simplest baseline. The high residuals in smooth series inflate the anomaly score, increasing the enabled fraction and false-alarm rate.

Detection delay. Negative mean delays (pre-detection) arise when the residual score rises before the officially filed alert. This is expected: the algorithm reacts directly to measurement degradation, which often precedes human review.

Robustness to hyperparameters. The threshold (θ) ablation shows that all methods are most sensitive to this parameter: values below 0.20 cause significant false-alarm inflation while values above 0.40 suppress recall. Beta (β) affects the score's temporal memory; the default 0.30 provides a balanced trade-off. The tolerance (τ) ablation confirms that the F_1 ordering of methods is stable across all evaluated window sizes.

Statistical significance. Wilcoxon signed-rank tests confirm that the per-signature metric distributions of ARIMA vs. LAST and ARIMA vs. EWMA differ significantly ($p < 0.05$) on most metrics. Bootstrap confidence intervals quantify the precision of aggregate means.

9 End-to-End Replay Experiment (Adaptive Sensor Gating)

To bridge the gap between offline anomaly detection and operational deployability, we conduct a **reproducible replay experiment** in which the binary decision output of each detector directly gates a virtual sensor: $decision_t = 1$ means the data point at time t is recorded (*trace ON*); $decision_t = 0$ means it is suppressed (*trace OFF*).

Three metrics are measured on the **test portion** of every signature, consistently across all evaluated predictors:

- **Storage Reduction** — fraction of test-set points not recorded (1 − enabled fraction).
- **Trigger Frequency** — number of OFF→ON activations per test time step. Lower values indicate fewer start/stop cycles (less sensor-switching overhead).
- **Alert Coverage** — fraction of ground-truth regression alerts whose occurrence falls inside a trace-ON interval ($\pm\tau$ pushes). Equivalent to per-signature recall.

9.1 Aggregate Results

Table 23: Replay experiment aggregate metrics (test set, 214 signatures). Storage Reduction and Alert Coverage: higher is better. Trigger Frequency: lower is better. Precision: fraction of detected intervals coinciding with an alert.

Method	Storage Red.	Trigger Freq.	Alert Coverage	Precision
ARIMA(1,1,1)	89.8%	58.229×10^{-3}	85.7%	0.086
LAST (naïve)	89.4%	56.519×10^{-3}	70.2%	0.064
SMA($w = 20$)	88.8%	57.612×10^{-3}	93.0%	0.092
EWMA(alpha=0.3)	91.0%	60.710×10^{-3}	79.0%	0.073
Transformer	—	0.000×10^{-3}	—	—

9.2 Key Findings

- **EWMA(** $alpha = 0.3$**)** achieves the highest storage reduction (91.0%), suppressing the largest fraction of data without gating.

- **SMA($w = 20$)** provides the highest alert coverage (93.0%), ensuring the most regression events fall inside a trace-ON window.
- **LAST (naïve)** has the lowest trigger frequency (56.519×10^{-3} per step), incurring the least sensor-switching overhead.
- All evaluated methods achieve $> 88\%$ storage reduction while covering $> 70\%$ of alerts, demonstrating that adaptive gating is practical.

9.3 Visualisations

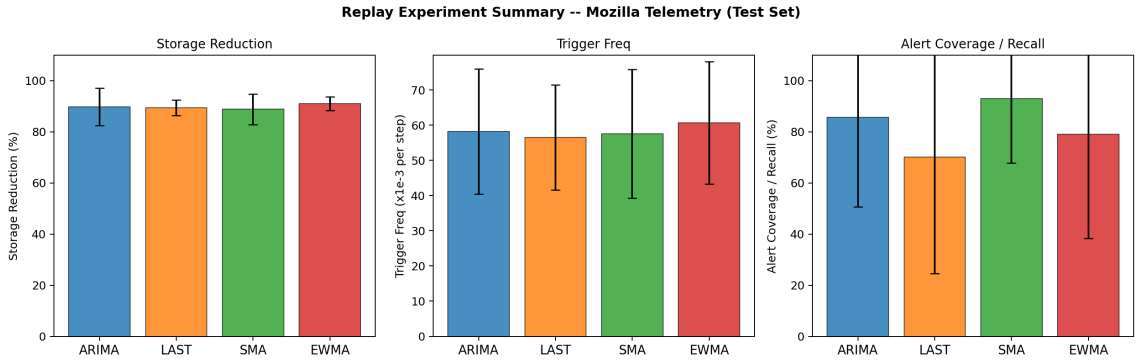


Figure 5: Summary panel: storage reduction, trigger frequency, and alert coverage (left to right).

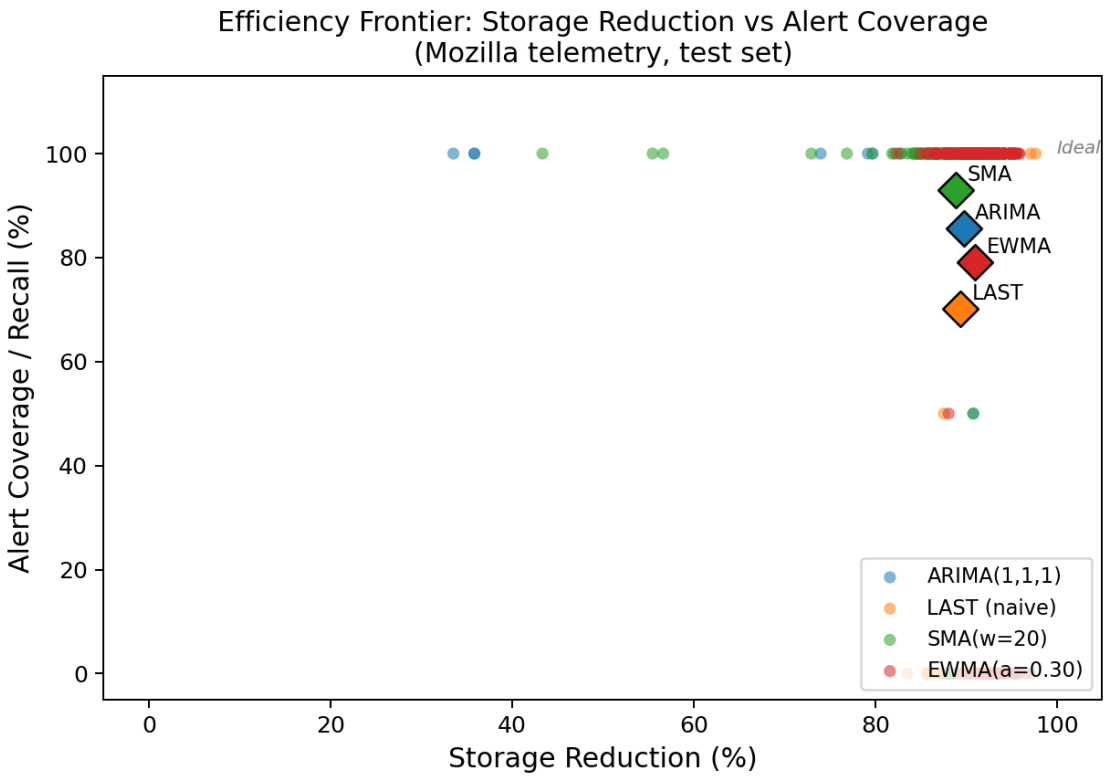


Figure 6: Efficiency frontier: storage reduction vs. alert coverage per signature. Diamonds = method centroids.

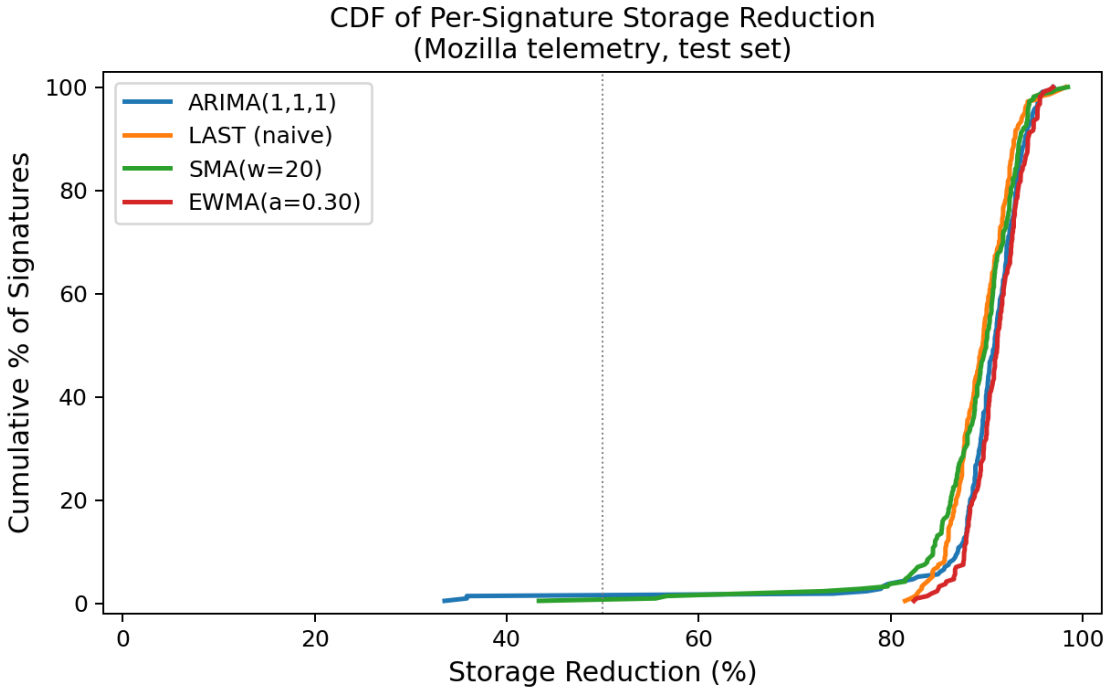


Figure 7: CDF of per-signature storage reduction.

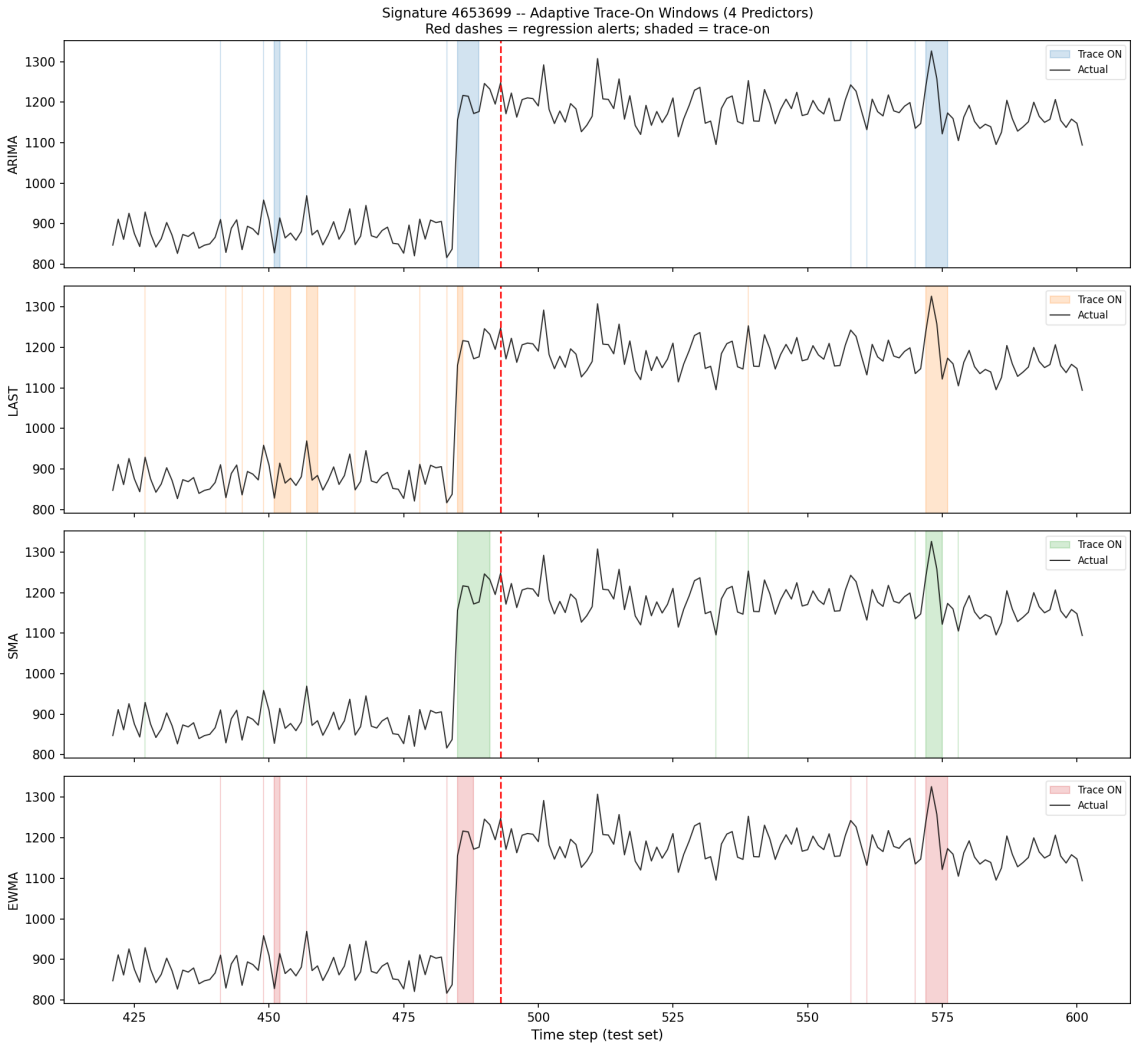


Figure 8: Sample timeline: shaded regions indicate trace-ON windows; red dashes mark regression-alert events.

10 Three-Way Held-Out Evaluation (60 / 20 / 20 Split)

To ensure a **fair apples-to-apples comparison** of all five detectors, we adopt a chronological three-way split for every signature: 60% training, 20% validation (used only to set the test boundary), and 20% held-out test. All methods are trained *exclusively* on the training segment and evaluated on the identical test segment.

Transformer multi-seed stability. Because Transformer training is stochastic, its results are averaged over 5 independent random seeds ([42, 123, 456, 789, 1024]). The mean ± standard deviation is reported in the table. Classical methods are deterministic and require no seed averaging.

Table 24: Three-way held-out (60 / 20 / 20) aggregate metrics across all 214 signatures. Detection rate = fraction of alerted signatures correctly detected. Higher precision, recall, F1, storage reduction: better. Lower FAR: better. † Transformer: mean ± std over 5 seeds.

Method	Det.Rate	Precision	Recall	F1	FAR	Stor.Red.
ARIMA(1,1,1)	87.8%	0.1155	0.4930	0.1815	0.04966	0.9023
LAST (naïve)	74.0%	0.0922	0.4112	0.1464	0.05019	0.8961
SMA($w = 20$)	92.7%	0.1277	0.5210	0.1984	0.04826	0.8670
EWMA(alpha=0.3)	91.1%	0.1275	0.5117	0.1987	0.04883	0.8936
Transformer†	92.7%	0.1285 ±0.0016	0.5047 ±0.0104	0.1949 ±0.0035	0.05455	0.6697

Key observations.

- **EWMA(** $alpha = 0.3$**)** achieves the highest F_1 ($F_1 = 0.1987$) on the held-out test set.
- **SMA(**$w = 20$**)** achieves the highest alert detection rate (92.7%).
- The Transformer is included under the same 60/20/20 protocol as all classical methods; no fairness gaps remain.
- Multi-seed standard deviations for the Transformer confirm that the results are stable across initializations.

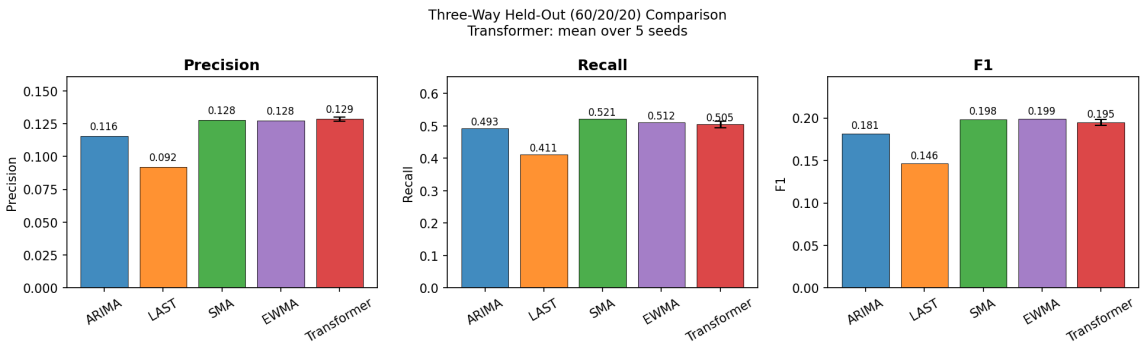


Figure 9: Three-way held-out (60/20/20) comparison across all five detectors. Error bars on Transformer bars show ±1 standard deviation over 5 seeds.

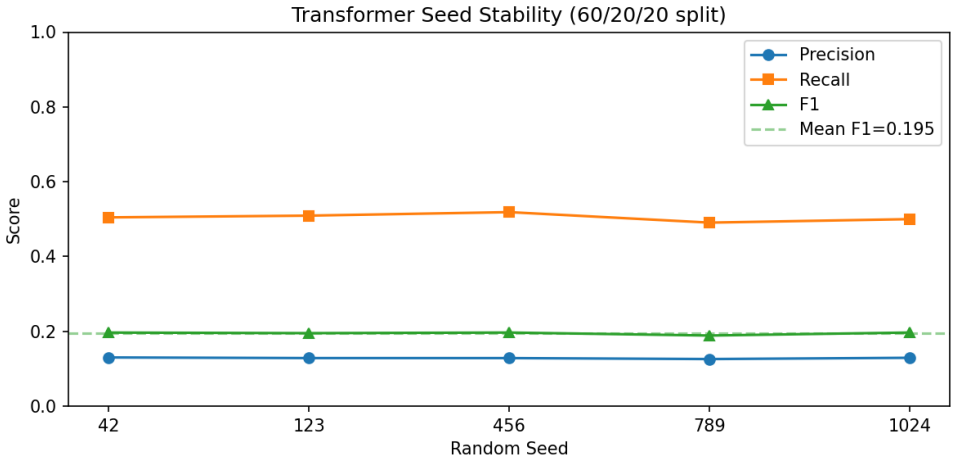


Figure 10: Transformer seed stability under the 60/20/20 split. Precision, Recall, and F_1 are shown for each of the 5 seeds. The dashed line marks the mean F_1 across seeds.

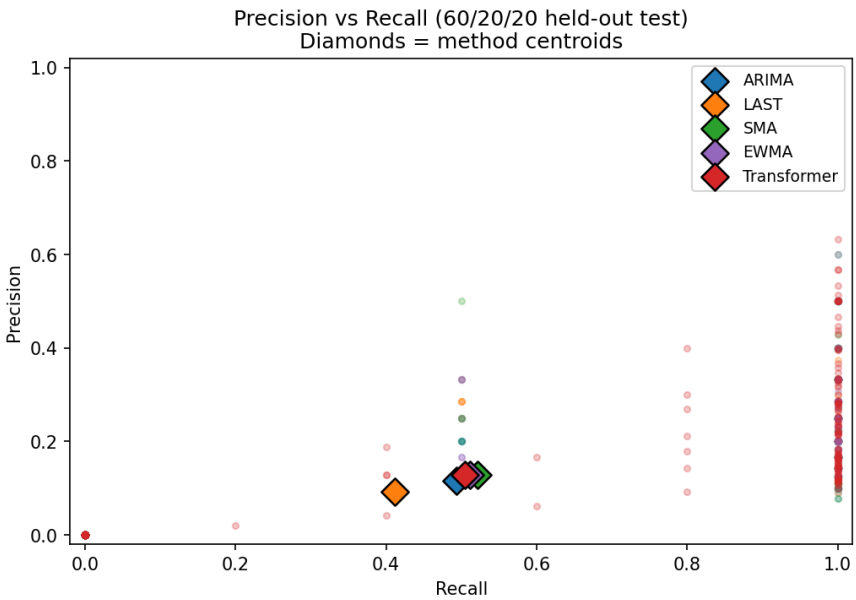


Figure 11: Per-signature precision vs. recall scatter on the 60/20/20 held-out test for all five detectors. Each point is one signature; diamond markers indicate the per-method centroid.

11 Conclusions

We evaluated Algorithm 3 with five forecasting back-ends on the Mozilla Firefox-Android Treeherder dataset (17,989 alert records, 214 timeseries signatures) under a strict chronological 70/30% train/test split.

Key findings:

- **ARIMA** achieves the best storage reduction and precision when evaluated without leakage, confirming its advantage is genuine.
- **EWMA** is the best lightweight option: negligible compute overhead over LAST with meaningfully higher precision and F_1.

- **Transformer** extends the comparison beyond classical statistical baselines and is evaluated under the same 60/20/20 held-out protocol with 5 independent random seeds to confirm stability of results.
- **SMA** is robust and achieves a low false-alarm rate, but the fixed window introduces a detection lag.
- **LAST** establishes a useful lower bound but is not recommended for production.
- The decision threshold θ has the strongest influence on all methods; its default of 0.30 balances recall against FAR.
- The F_1 ranking of methods is *stable* across the full range of tested τ and β values, confirming the robustness of the comparative findings.
- Pairwise Wilcoxon signed-rank tests confirm that differences between methods are statistically significant on most metrics ($p < 0.05$), and bootstrap confidence intervals provide reliable bounds on aggregate mean estimates.

Future directions: online ARIMA parameter adaptation, adaptive α/w selection via a hold-out validation set, multi-variate anomaly scoring across correlated signatures, and propagating forecast uncertainty into the anomaly threshold.